Daniel H. Lee

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LinkedIn | GitHub | Website

Education

Northeastern University Boston, MA
College of Science, MS in Bioinformatics | **GPA: 4.0** September 2025 –Expected December 2027
Khoury College of Computer Sciences, BS in Computer Science September 2020 – December 2024

Projects

Cross-Disorder Psychiatric GWAS Clustering | Independent Research | GitHub April 2026 – Present

- Extended Grotzinger et al. (Nature, 2026) cross-disorder Genomic SEM framework to investigate locus-level subtypes of pleiotropy across five psychiatric genomic factors (compulsive, schizophrenia–bipolar, neurodevelopmental, internalizing, substance use), using PGC summary statistics spanning 14 disorders
- Standardized factor-level effect sizes via z-scores and applied hierarchical clustering as the primary analysis, benchmarked against k-means, Gaussian mixture models, and DBSCAN; selected k=3 (n=611 loci) based on silhouette scores and bootstrap stability
- Validated clusters using Q_P heterogeneity statistics, demonstrating a monotonic gradient across all five factors (Kruskal–Wallis $p < 0.001$) and significant enrichment of heterogeneous loci in the most factor-specific cluster (chi-squared $p = 7e-54$)
- Identified a 74-locus cluster showing antagonistic pleiotropy (schizophrenia–bipolar negative, internalizing positive) with significant enrichment for cell-cell adhesion and cadherin signaling pathways (g:Profiler, GO:BP and KEGG)

LUAD Metastatic Site Classification | Course Project, with Jason Chen | GitHub January 2026 – April 2026

- Co-developed a multi-label ClassifierChain pipeline to predict metastatic organotropism across 7 sites (MSK Cancer Cell 2023, cBioPortal), comparing clinical, genomic, and combined feature sets across linear and nonlinear models
- Built the genomic feature pipeline and PyTorch MLP with per-label positive weighting for severe class imbalance; identified that linear models with L2 regularization generalized best, while MLP showed severe overfitting (train–test F1 gap +0.546) at this sample size
- Conducted a demographic fairness audit across sex and race subgroups, finding no meaningful sex disparity and surfacing cohort limitations (84.8% White) that preclude racial fairness evaluation without multi-institutional data

Experience

MGH Oral and Maxillofacial Surgery | Medical Scribe Boston, MA | July 2025 –February 2026

- Documented surgical procedures, diagnostic imaging findings, and treatment plans in real-time EHR systems across oncology, reconstructive, and trauma surgery, developing fluency with clinical data workflows and medical terminology.
- Supported multi-disciplinary care coordination by synthesizing complex patient histories into structured clinical documentation, gaining firsthand exposure to clinical decision-making processes

Fidelity Investments | Software Engineer Co-op Boston, MA | July 2023 – December 2023

- Engineered an automated Python-based data extraction pipeline within Fidelity's Asset Management Software team, parsing and analyzing financial reports with automated validation checks that reduced manual review time by several hours daily for business users.
- Optimized large-scale file synchronization workflow across Amazon S3 and production databases using multithreading, reducing processing time for 100,000+ files from 40 minutes to 5 minutes (8x improvement)

Skills

Languages: Python, R, Bash, SQL
Statistical & Machine Learning Methods: Clustering (hierarchical, k-means, GMM, DBSCAN), classification (logistic regression, SVM, random forest, MLP), survival analysis, multi-label learning, feature importance analysis, scikit-learn, PyTorch
Bioinformatics & Infrastructure: GWAS summary statistics, pandas, BioPython, NGS analysis (DNA & RNA-seq), sequence alignment, data visualization, Docker, Git, HPC (SLURM)